

An energy company is considering an investment in wind-power production. To determine whether this investment is economically feasible, the company seeks to estimate the expected revenue of a wind-power producer operating in a competitive spot market.

Consider a day-ahead spot market with five electricity producers. Three producers generate electricity from conventional sources, such as coal, whereas the remaining two rely on solar and wind power, respectively. The market is cleared once each day for 24 hourly trading intervals of the following day. For each interval, every producer submits a bid that specifies the electricity quantity it is willing to supply (MWh), together with the minimum price it is willing to accept ($\$/\text{MWh}$). The market operator then applies a merit-order clearing mechanism to determine both the market-clearing price and the dispatch schedule, subject to the requirement that total electricity demand be satisfied. If a producer's bid is accepted, the producer is obligated to supply the nominated quantity and is remunerated at the market-clearing price. If the bid is not accepted, the producer receives no revenue for that interval.

The energy company intends to develop a model to estimate the expected revenue of the wind-power producer on any future day by determining the hourly bid quantities and prices under uncertainty. On this basis, the model can subsequently be extended to examine outcomes across a range of short-term and long-term scenarios, thereby supporting an assessment of the investment's economic viability.

Please formalise this problem as an agent-based model, and specify the key components, including time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables. At this stage, the model should be limited to simulating single-day market operations.

Show less

Agent-Based Model: Day-Ahead Wind-Power Revenue Estimation

1. Time Scales

The model operates across two nested time scales.

Scheduling horizon: A single calendar day, subdivided into $T = 24$ sequential hourly trading intervals, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$.

Market clearing: The day-ahead auction is cleared once, prior to real-time delivery, simultaneously for all 24 intervals. Bidding and clearing therefore precede physical dispatch; there is no intra-day re-trading in this single-day formulation.

2. Agents

Five producers participate as strategic (or price-taking) agents, each submitting one price-quantity bid per interval.

Agent ID	Technology	Bid behaviour
C_1, C_2, C_3	Conventional (coal)	Deterministic availability; bids reflect marginal fuel cost
S	Solar PV	Stochastic output; zero marginal cost
W	Wind	Stochastic output; zero marginal cost — the agent of primary interest

A sixth, passive entity — the **Market Operator** — clears the market via the merit-order mechanism but makes no strategic decisions and holds no objective of its own.

3. State Variables

State variables characterise the system's condition at each interval t .

Market-level states

Symbol	Description	Domain
D_t	Aggregate electricity demand	$\mathbb{R}_{>0}$ (MWh)
λ_t	Market-clearing price	$\mathbb{R}_{\geq 0}$ (\\$/MWh)
$\mathcal{A}_t$	Set of accepted bids at interval t	$\subseteq \{C_1, C_2, C_3, S, W\}$

Agent-level states

Symbol	Description	Domain
q_t^i	Dispatched quantity of agent i	$[0, \bar{Q}_t^i]$ (MWh)
r_t^i	Revenue earned by agent i at interval t	$\mathbb{R}_{\geq 0}$ (\\$)
$\hat{Q}_t^W$	Wind producer's realised available capacity	$[0, \bar{Q}^W]$ (MWh)
$\hat{Q}_t^S$	Solar producer's realised available capacity	$[0, \bar{Q}^S]$ (MWh)

4. Control Variables

Control variables are the decision instruments available to each agent prior to market clearing.

For each agent $i \in \{C_1, C_2, C_3, S, W\}$ and each interval t :

Symbol	Description	Domain
b_t^i	Bid price (minimum acceptable price)	$[\underline{p}, \bar{p}]$ (\\$/MWh)
v_t^i	Bid quantity (offered supply)	$[0, \bar{Q}_t^i]$ (MWh)

The wind producer's **primary control variables** are therefore the pair (b_t^W, v_t^W) for each $t \in \mathcal{T}$, chosen before $\hat{Q}_t^W$ is observed, under uncertainty about both demand and competitor bids.

5. Processes

5.1 Capacity Realisation

Before clearing, each stochastic producer's available capacity is drawn from its respective distribution (see §8). Conventional producers are assumed to operate at a fixed, known capacity within the day.

5.2 Bid Submission

Each agent submits the pair (b_t^i, v_t^i) for all t . Conventional producers may bid at or near their short-run marginal cost; renewable producers typically bid at zero or a small positive price, with quantity capped at realised available capacity.

5.3 Merit-Order Market Clearing

For each interval t , the market operator:

1. Ranks all submitted bids in ascending order of bid price to form the **aggregate supply curve**.
2. Accepts bids sequentially from cheapest to most expensive until cumulative accepted quantity meets or exceeds D_t .

3. Sets the **market-clearing price** λ_t equal to the bid price of the **marginal (last-accepted) producer**.
4. Dispatches every accepted producer at their bid quantity v_t^i and remunerates them at λ_t .

Formally, define the merit-order ranking σ_t such that $b_t^{\sigma_t(1)} \leq b_t^{\sigma_t(2)} \leq \dots \leq b_t^{\sigma_t(5)}$. Then:

$$\lambda_t = b_t^{\sigma_t(k^*)}, \quad \text{where } k^* = \min \left\{ k : \sum_{j=1}^k v_t^{\sigma_t(j)} \geq D_t \right\}$$

5.4 Revenue Calculation

Agent i 's interval revenue is:

$$r_t^i = \begin{cases} \lambda_t \cdot v_t^i & \text{if } i \in \mathcal{A}_t \\ 0 & \text{otherwise} \end{cases}$$

5.5 Daily Revenue Aggregation

The wind producer's total daily revenue is:

$$R^W = \sum_{t=1}^{24} r_t^W$$

The model's primary output is the **expected daily revenue** $\mathbb{E}[R^W]$, estimated via Monte Carlo simulation over realisations of all stochastic variables.

6. Objectives

Wind producer W seeks to maximise expected daily revenue:

$$\max_{\{b_t^W, v_t^W\}_{t=1}^{24}} \mathbb{E} \left[\sum_{t=1}^{24} r_t^W \right]$$

subject to:

$$v_t^W \leq \hat{Q}_t^W \quad \forall t, \quad b_t^W \in [\underline{p}, \bar{p}], \quad v_t^W \geq 0$$

Conventional producers are modelled as price-takers bidding at short-run marginal cost, and **solar** is modelled analogously to wind. In the baseline formulation, competitor strategies are treated as exogenous inputs rather than co-optimised objectives, which reduces the model to a **single-agent optimisation under uncertainty**.

7. Constants

Symbol	Description
$T = 24$	Number of trading intervals per day
$\bar{Q}^{C_k}$	Installed capacity of conventional producer $k \in \{1, 2, 3\}$ (MWh)
$\bar{Q}^W, \bar{Q}^S$	Nameplate capacity of wind and solar producers (MWh)
mc^{C_k}	Short-run marginal cost of conventional producer k (\\$/MWh)
$\underline{p}, \bar{p}$	Market-wide bid price floor and cap (\\$/MWh)

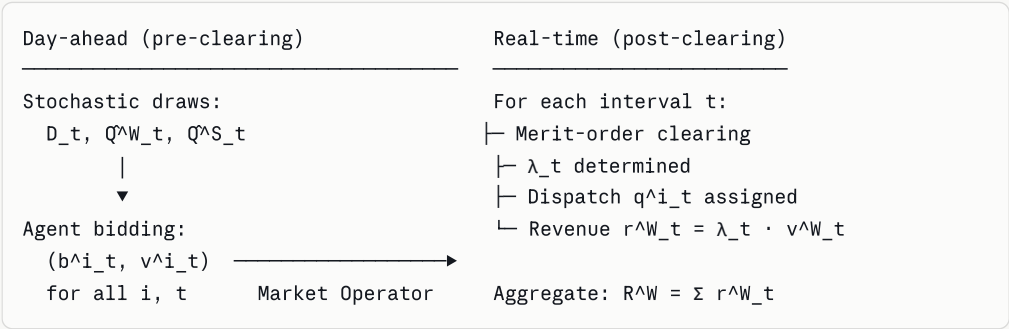
8. Uncertain Parameters and Stochastic Variables

These quantities are unknown at bid submission time and are modelled probabilistically.

Symbol	Description	Suggested model
D_t	Hourly demand	Log-normal or empirical distribution; time-of-day seasonality; autocorrelated across t
$\hat{Q}_t^W$	Wind available capacity	Beta or Weibull-derived distribution on $[0, \bar{Q}^W]$; autocorrelated; driven by wind-speed forecast error
$\hat{Q}_t^S$	Solar available capacity	Deterministic diurnal envelope scaled by Beta-distributed cloud-cover factor; zero outside daylight hours
$b_t^{C_k}$	Rival conventional bid prices	Modelled as fixed at mc^{C_k} in baseline; can be perturbed by a noise term for sensitivity analysis
ϵ_t^W	Wind forecast error	Zero-mean additive process; captures deviation between forecast used for bidding and realised output

Key dependency structure: D_t and $\hat{Q}_t^W$ may be negatively correlated (high wind often accompanies low-demand weather conditions), and this covariance materially affects revenue estimation. It should be captured explicitly, for example via a joint copula or a shared meteorological driver.

Summary Schematic



This specification defines a fully simulable single-day model. Natural extensions include multi-day simulation with learning, strategic competitor behaviour via Nash equilibrium bidding, and long-run capacity investment analysis — all of which build directly on the components formalised here.

You can assume that the market demand follows a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

8. Uncertain Parameters and Stochastic Variables *(Revised)*

These quantities are unknown at bid submission time and are modelled probabilistically.

Symbol	Description	Suggested model
D_t	Hourly demand	Normal distribution with constant mean and variance (see below)
$\hat{Q}_t^W$	Wind available capacity	Beta or Weibull-derived distribution on $[0, \bar{Q}^W]$; autocorrelated; driven by wind-speed forecast error
$\hat{Q}_t^S$	Solar available capacity	Deterministic diurnal envelope scaled by Beta-distributed cloud-cover factor; zero outside daylight hours
$b_t^{C_k}$	Rival conventional bid prices	Fixed at mc^{C_k} in baseline; perturbed by additive noise term for sensitivity analysis
ϵ_t^W	Wind forecast error	Zero-mean additive process; captures deviation between forecast used for bidding and realised output

Demand Process

Hourly demand is modelled as an independent normal random variable, identically distributed across all intervals:

$$D_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_D, \sigma_D^2), \quad \forall t \in \mathcal{T}$$

where $\mu_D > 0$ and $\sigma_D^2 > 0$ are fixed constants throughout the simulation horizon.

Two practical constraints follow from this specification. First, since physical demand cannot be negative, realisations $D_t \leq 0$ must be handled explicitly. Provided that $\mu_D \gg \sigma_D$ — a condition that holds under any reasonable calibration — the probability of a non-positive draw is negligible, but it is nonetheless recommended to enforce $D_t = \max(D_t, \underline{D})$ for a small positive floor $\underline{D}$ to preserve physical admissibility. Second, the assumption of a **constant mean and variance** across all 24 intervals abstracts away intra-day demand variation (e.g. morning ramp, evening peak). This is appropriate for an initial feasibility model; the specification can be relaxed in later iterations by replacing μ_D and σ_D^2 with hour-indexed counterparts $\mu_{D,t}$ and $\sigma_{D,t}^2$ without altering any other component of the model.

Revised Dependency Structure

Under this specification, demand draws are independent across intervals and, by assumption, independent of $\hat{Q}_t^W$ and $\hat{Q}_t^S$. This simplifies Monte Carlo estimation considerably, as joint draws reduce to drawing each D_t marginally. If empirical evidence later suggests a meaningful negative correlation between wind output and demand — a common feature of real markets — the constant-variance normal model can be extended to a bivariate normal, retaining analytical tractability while restoring the dependency.

You can also make the following assumptions regarding the behavioural logic of the producers: (1) conventional producers have stable bid quantities, with bid prices following normal distributions with uncertain means and variances; (2) the solar-power producer's bid quantity follows $\max(0, a + b \cos(\frac{2\pi t}{24}))$, with bid prices following a normal distribution with uncertain mean and variance; and (3) actual wind-power production is stochastic, following a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

4. Control Variables (Revised)

Control variables are the decision instruments submitted by each agent prior to market clearing. The assumptions below specify the structure of bid quantities and prices for each producer class.

4.1 Conventional Producers ($C_k, k = 1, 2, 3$)

Bid quantities are **stable**, taken as fixed constants across all intervals:

$$v_t^{C_k} = \bar{Q}^{C_k}, \quad \forall t \in \mathcal{T}$$

Bid prices are drawn from producer-specific normal distributions whose means and variances are themselves uncertain parameters (see §8):

$$b_t^{C_k} \sim \mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k}), \quad \forall t \in \mathcal{T}$$

Bid prices are drawn independently across intervals and across producers.

4.2 Solar Producer (S)

Bid quantity follows a deterministic cosine profile reflecting the diurnal solar cycle, truncated below at zero:

$$v_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad \forall t \in \mathcal{T}$$

where a is a baseline offset and $b > 0$ is the amplitude. The truncation ensures physical admissibility: the quantity is zero during night-time intervals where the cosine term drives the expression negative. Bid prices are drawn from a normal distribution:

$$b_t^S \sim \mathcal{N}(\mu_b^S, \sigma_b^{2,S}), \quad \forall t \in \mathcal{T}$$

4.3 Wind Producer (W)

Actual wind-power production is stochastic and constitutes the operative constraint on the bid quantity. Realisations are drawn from:

$$\hat{Q}_t^W \sim \mathcal{N}(\mu_W, \sigma_W^2), \quad \forall t \in \mathcal{T}$$

with physical admissibility enforced via truncation, $\hat{Q}_t^W = \max(0, \hat{Q}_t^W)$. The wind producer's bid quantity is set equal to realised production — the quantity it is able and obligated to deliver — and bid price is fixed at zero, reflecting zero marginal cost and a strategy of always clearing the market:

$$v_t^W = \hat{Q}_t^W, \quad b_t^W = 0, \quad \forall t \in \mathcal{T}$$

Under this price-taking strategy, the wind producer's control problem reduces to characterising the **distribution of revenue** as a function of the stochastic variables $\hat{Q}_t^W, D_t$, and competitor bids, rather than optimising over (b_t^W, v_t^W) .

8. Uncertain Parameters and Stochastic Variables (*Revised*)

Symbol	Description	Distribution
D_t	Hourly market demand	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
$\hat{Q}_t^W$	Wind producer realised output	$\mathcal{N}(\mu_W, \sigma_W^2)$, i.i.d. across t ; truncated below at 0
$b_t^{C_k}$	Conventional producer k bid price	$\mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$, i.i.d. across t
b_t^S	Solar producer bid price	$\mathcal{N}(\mu_b^S, \sigma_b^{2,S})$, i.i.d. across t
$\mu_b^{C_k}, \sigma_b^{2,C_k}$	Mean and variance of conventional producer k 's bid price	Uncertain constants; estimated from historical data or elicited as prior distributions
$\mu_b^S, \sigma_b^{2,S}$	Mean and variance of solar producer's bid price	Uncertain constants; estimated from historical data or elicited as prior distributions

Notes on the Stochastic Structure

Demand. As previously specified, $D_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_D, \sigma_D^2)$ with a floor $\underline{D} > 0$ enforced to maintain physical admissibility.

Wind output. The assumption $\hat{Q}_t^W \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_W, \sigma_W^2)$ treats production as constant in expectation across all 24 intervals, which abstracts away intra-day wind-speed variation. This is consistent with the constant-mean demand assumption and appropriate for a first-pass feasibility model. The truncation $\hat{Q}_t^W \leftarrow \max(0, \hat{Q}_t^W)$ introduces a slight upward bias in the effective mean; if $\mu_W \gg \sigma_W$, this bias is negligible. An upper truncation at $\bar{Q}^W$ should additionally be imposed to respect nameplate capacity.

Conventional bid prices. The means $\mu_b^{C_k}$ are conceptually anchored to each producer's short-run marginal cost, but are treated as uncertain to reflect incomplete information about rivals' cost structures and bidding strategies. The variances σ_b^{2,C_k} capture strategic or behavioural randomness around those costs.

Solar bid price. The solar producer's quantity is fully deterministic given parameters a and b , so price is the sole source of stochasticity in its bid. Like wind, solar has near-zero marginal cost; the positive mean μ_b^S may reflect a small strategic markup.

Independence assumptions. All stochastic variables are assumed mutually independent across agents and across intervals. This precludes, for example, correlation between wind output and demand, or between competitor bid prices and prevailing conditions. These are

simplifying assumptions appropriate for the initial model that may be relaxed in subsequent iterations.

You should not make any assumptions about the wind-power producer's bid quantity and price, as these are determined by the users. Please refine your specification accordingly and present the revised section.

4. Control Variables (*Revised*)

4.3 Wind Producer (W)

Actual wind-power production is stochastic, constraining the maximum quantity the wind producer can physically deliver. Realisations are drawn from:

$$\hat{Q}_t^W \sim \mathcal{N}(\mu_W, \sigma_W^2), \quad \forall t \in \mathcal{T}$$

with physical admissibility enforced via truncation at zero and nameplate capacity:

$$\hat{Q}_t^W \leftarrow \min(\bar{Q}^W, \max(0, \hat{Q}_t^W))$$

Unlike the other producers, the wind producer's **bid quantity and bid price are free decision variables**, specified exogenously by the user for each interval:

$$v_t^W \in [0, \hat{Q}_t^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

The bid quantity is bounded above by realised production, as the producer cannot offer more electricity than it is physically capable of delivering. No assumptions are imposed on the user's chosen values of v_t^W and b_t^W ; these constitute the **primary inputs** to the model, and the simulation is conditioned on them. The model's purpose is then to characterise the **distribution of revenue** $R^W = \sum_{t=1}^{24} r_t^W$ that results from any user-specified bidding strategy $(b_t^W, v_t^W)_{t=1}^{24}$, under uncertainty over demand, wind output, and competitor bids.

This formulation supports two distinct modes of use. In **evaluation mode**, the user supplies a fixed bidding strategy and the model returns the resulting expected revenue and its distribution. In **optimisation mode**, the user may search over the space of admissible strategies to identify the pair (b_t^W, v_t^W) that maximises expected revenue or some other objective, with the model serving as the simulation oracle for each candidate strategy.

When multiple bids at the clearing price collectively exceed the residual demand, producers are dispatched in priority order: solar $\rightarrow$ wind $\rightarrow$ conv1 $\rightarrow$ conv2 $\rightarrow$ conv3, with the last accepted producer potentially dispatched only partially to exactly meet demand. Please refine your specification accordingly and present the revised section.

5. Processes (*Revised*)

5.3 Merit-Order Market Clearing

For each interval t , the market operator clears the market as follows.

Step 1 — Rank bids. Sort all submitted bids in ascending order of bid price. Let σ_t denote this ranking, so that $b_t^{\sigma_t(1)} \leq b_t^{\sigma_t(2)} \leq \dots \leq b_t^{\sigma_t(5)}$.

Step 2 — Identify the marginal price. Accepting bids sequentially in merit order, find the smallest set of bids whose cumulative offered quantity meets or exceeds demand:

$$k^* = \min \left\{ k : \sum_{j=1}^k v_t^{\sigma_t(j)} \geq D_t \right\}$$

The **market-clearing price** is set at the bid price of the last accepted producer in this unconstrained ranking:

$$\lambda_t = b_t^{\sigma_t(k^*)}$$

Step 3 — Resolve ties at the marginal price. Let $\mathcal{M}_t$ denote the set of producers whose bid price equals λ_t , and define the **residual demand** at the marginal price as the demand not yet satisfied by all strictly infra-marginal producers:

$$D_t^{\text{res}} = D_t - \sum_{\{j : b_t^j < \lambda_t\}} v_t^j$$

If the aggregate quantity offered by $\mathcal{M}_t$ exceeds D_t^{res} , producers in $\mathcal{M}_t$ are dispatched according to the fixed priority order:

$$S \succ W \succ C_1 \succ C_2 \succ C_3$$

Producers are accepted in this priority sequence until D_t^{res} is exactly met. Formally, let $\pi = (\pi_1, \pi_2, \dots)$ be the subsequence of $\mathcal{M}_t$ ordered by priority. The dispatched quantities for producers in $\mathcal{M}_t$ are:

$$q_t^{\pi_j} = \begin{cases} v_t^{\pi_j} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \\ D_t^{\text{res}} - \sum_{l=1}^{j-1} v_t^{\pi_l} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \leq \sum_{l=1}^j v_t^{\pi_l} \\ 0 & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} \geq D_t^{\text{res}} \end{cases}$$

The **last accepted producer** π_{j^*} — where j^* is the index at which cumulative accepted quantity first reaches D_t^{res} — may therefore be **partially dispatched**, supplying only the residual quantity required to exactly satisfy demand.

Step 4 — Assemble full dispatch. The dispatched quantity for every producer i is:

$$q_t^i = \begin{cases} v_t^i & \text{if } b_t^i < \lambda_t \quad (\text{strictly infra-marginal}) \\ q_t^i \text{ from Step 3} & \text{if } b_t^i = \lambda_t \quad (\text{marginal, resolved by priority}) \\ 0 & \text{if } b_t^i > \lambda_t \quad (\text{supra-marginal}) \end{cases}$$

Step 5 — Compute revenues. Every accepted producer — including any partially dispatched producer — is remunerated at the uniform clearing price λ_t :

$$r_t^i = \lambda_t \cdot q_t^i, \quad \forall i$$

Remark on Priority and Wind Revenue

The priority rule $S \succ W \succ C_1 \succ C_2 \succ C_3$ favours renewable producers, reflecting standard renewable dispatch preference. For the wind producer specifically, this rule is consequential: whenever wind's bid price ties with one or more conventional producers at the marginal price, wind is dispatched ahead of all three conventional producers but behind solar. This creates an asymmetry in partial-dispatch risk — wind bears that risk only when solar's offered quantity alone is insufficient to cover D_t^{res} and wind is the marginal producer in the priority sequence. This interaction should be accounted for when interpreting the distribution of R^W across simulation runs.

Given that actual wind-power production is stochastic, it may fall short of the quantity committed to dispatch one day in advance. If the wind-power producer under-delivers relative to its dispatched quantity, the shortfall incurs a constant penalty cost (\$/MWh). Please incorporate this penalty mechanism into your specification.

Then, please provide a complete model specification that specifies all key components, including the time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables.

Agent-Based Model: Day-Ahead Wind-Power Revenue Estimation

Complete Specification

1. Time Scales

The model operates across two nested time scales.

Scheduling horizon: A single calendar day, subdivided into $T = 24$ sequential hourly trading intervals, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$.

Market clearing: The day-ahead auction is cleared once, simultaneously for all 24 intervals, prior to real-time delivery. Bid submission and market clearing therefore precede physical dispatch; committed quantities become binding obligations subject to penalty upon under-delivery.

2. Agents

Five producers participate as agents, each submitting one price-quantity bid per interval. A sixth, passive entity — the **Market Operator** — clears the market via the merit-order mechanism but makes no strategic decisions.

Agent ID	Technology	Bid behaviour
C_1, C_2, C_3	Conventional (coal)	Stable quantities; stochastic bid prices
S	Solar PV	Deterministic cosine quantity profile; stochastic bid price
W	Wind	User-specified bid quantity and price; stochastic realised output

3. State Variables

State variables characterise the condition of the system at each interval t .

Market-level states

Symbol	Description	Domain
D_t	Aggregate electricity demand	$\mathbb{R}_{>0}$ (MWh)
λ_t	Market-clearing price	$\mathbb{R}_{\geq 0}$ (\\$/MWh)
D_t^{res}	Residual demand at the marginal price	$\mathbb{R}_{\geq 0}$ (MWh)

Agent-level states

Symbol	Description	Domain
q_t^i	Dispatched quantity of agent i at interval t	$[0, v_t^i]$ (MWh)
$\hat{Q}_t^W$	Wind producer's realised output at interval t	$[0, \bar{Q}^W]$ (MWh)
δ_t^W	Wind producer's shortfall at interval t	$\mathbb{R}_{\geq 0}$ (MWh)
r_t^i	Revenue of agent i at interval t	$\mathbb{R}$ (\\$)

4. Control Variables

Control variables are the decision instruments submitted by each agent prior to market clearing.

4.1 Conventional Producers ($C_k, k = 1, 2, 3$)

Bid quantities are stable constants across all intervals:

$$v_t^{C_k} = \bar{Q}^{C_k}, \quad \forall t \in \mathcal{T}$$

Bid prices are drawn independently from producer-specific normal distributions:

$$b_t^{C_k} \sim \mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k}), \quad \forall t \in \mathcal{T}$$

4.2 Solar Producer (S)

Bid quantity follows a deterministic cosine profile reflecting the diurnal solar cycle, truncated below at zero:

$$v_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad \forall t \in \mathcal{T}$$

where a is a baseline offset and $b > 0$ is the amplitude. Bid prices are drawn independently from a normal distribution:

$$b_t^S \sim \mathcal{N}(\mu_b^S, \sigma_b^{2,S}), \quad \forall t \in \mathcal{T}$$

4.3 Wind Producer (W)

Actual wind-power production is stochastic. Realisations are drawn from a normal distribution and truncated to be physically admissible:

$$\hat{Q}_t^W \sim \mathcal{N}(\mu_W, \sigma_W^2), \quad \hat{Q}_t^W \leftarrow \min\left(\bar{Q}^W, \max(0, \hat{Q}_t^W)\right), \quad \forall t \in \mathcal{T}$$

The wind producer's bid quantity and bid price are **free decision variables specified exogenously by the user** for each interval:

$$v_t^W \in [0, \bar{Q}^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

These constitute the **primary inputs** to the model. The bid quantity v_t^W represents the quantity committed one day in advance, prior to observing $\hat{Q}_t^W$, and is not required to equal realised output. If v_t^W is accepted and $\hat{Q}_t^W < q_t^W$, the producer under-delivers and incurs a penalty (see §5.4).

5. Processes

5.1 Stochastic Realisations

At the start of each simulated day, the following quantities are drawn independently for each interval t :

- Demand: $D_t \sim \mathcal{N}(\mu_D, \sigma_D^2)$, floored at $\underline{D} > 0$
- Wind output: $\hat{Q}_t^W \sim \mathcal{N}(\mu_W, \sigma_W^2)$, truncated to $[0, \bar{Q}^W]$
- Conventional bid prices: $b_t^{C_k} \sim \mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$ for $k = 1, 2, 3$
- Solar bid price: $b_t^S \sim \mathcal{N}(\mu_b^S, \sigma_b^{2,S})$

Solar bid quantities v_t^S are computed deterministically from the cosine profile. The user-specified wind bids (v_t^W, b_t^W) are fixed inputs, not drawn stochastically.

5.2 Bid Submission

Each agent submits the pair (b_t^i, v_t^i) for all $t \in \mathcal{T}$ to the market operator before clearing.

5.3 Merit-Order Market Clearing

For each interval t , the market operator clears the market as follows.

Step 1 — Rank bids. Sort all submitted bids in ascending order of bid price. Let σ_t denote this ranking, so that $b_t^{\sigma_t(1)} \leq b_t^{\sigma_t(2)} \leq \dots \leq b_t^{\sigma_t(5)}$.

Step 2 — Identify the marginal price. Find the cheapest combination of bids whose cumulative offered quantity meets or exceeds demand:

$$k^* = \min \left\{ k : \sum_{j=1}^k v_t^{\sigma_t(j)} \geq D_t \right\}$$

The market-clearing price is set at the bid price of the last accepted producer in this unconstrained ranking:

$$\lambda_t = b_t^{\sigma_t(k^*)}$$

Step 3 — Resolve ties at the marginal price. Let $\mathcal{M}_t$ denote the set of producers whose bid price equals λ_t , and define the residual demand at the marginal price as:

$$D_t^{\text{res}} = D_t - \sum_{\{i : b_t^i < \lambda_t\}} v_t^i$$

If the aggregate quantity offered by $\mathcal{M}_t$ exceeds D_t^{res} , producers in $\mathcal{M}_t$ are dispatched according to the fixed priority order $S \succ W \succ C_1 \succ C_2 \succ C_3$. Let $\pi = (\pi_1, \pi_2, \dots)$ be the subsequence of $\mathcal{M}_t$ ordered by priority. Dispatched quantities are:

$$q_t^{\pi_j} = \begin{cases} v_t^{\pi_j} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \\ D_t^{\text{res}} - \sum_{l=1}^{j-1} v_t^{\pi_l} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \leq \sum_{l=1}^j v_t^{\pi_l} \\ 0 & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} \geq D_t^{\text{res}} \end{cases}$$

The last accepted producer π_{j^*} may be partially dispatched to exactly meet demand.

Step 4 — Assemble full dispatch. For every producer i :

$$q_t^i = \begin{cases} v_t^i & \text{if } b_t^i < \lambda_t \\ q_t^i \text{ from Step 3} & \text{if } b_t^i = \lambda_t \\ 0 & \text{if } b_t^i > \lambda_t \end{cases}$$

5.4 Shortfall Calculation and Penalty

After real-time delivery, the wind producer's realised output $\hat{Q}_t^W$ is compared against its dispatched commitment q_t^W . The shortfall is:

$$\delta_t^W = \max(0, q_t^W - \hat{Q}_t^W), \quad \forall t \in \mathcal{T}$$

A shortfall arises when $\hat{Q}_t^W < q_t^W$, i.e. realised production falls below the committed dispatch quantity. No penalty applies if the producer over-delivers ($\hat{Q}_t^W \geq q_t^W$); surplus output beyond q_t^W is uncompensated under this mechanism.

5.5 Interval Revenue

The net revenue of the wind producer at interval t combines market revenue from the accepted dispatch quantity and the penalty incurred by any shortfall:

$$r_t^W = \lambda_t \cdot q_t^W - \rho \cdot \delta_t^W, \quad \forall t \in \mathcal{T}$$

All other producers are remunerated at the clearing price without penalty:

$$r_t^i = \lambda_t \cdot q_t^i, \quad \forall i \neq W$$

5.6 Daily Revenue Aggregation

The wind producer's total net daily revenue is:

$$R^W = \sum_{t=1}^{24} r_t^W = \sum_{t=1}^{24} (\lambda_t \cdot q_t^W - \rho \cdot \delta_t^W)$$

The model's primary output is the **distribution of R^W** , and in particular its expectation $\mathbb{E}[R^W]$, estimated via Monte Carlo simulation over realisations of all stochastic variables.

6. Objective

The wind producer's objective is to select a bidding strategy $(b_t^W, v_t^W)_{t=1}^{24}$ that maximises expected net daily revenue:

$$\max_{\{b_t^W, v_t^W\}_{t=1}^{24}} \mathbb{E}[R^W] = \mathbb{E} \left[\sum_{t=1}^{24} (\lambda_t \cdot q_t^W - \rho \cdot \delta_t^W) \right]$$

subject to:

$$v_t^W \in [0, \bar{Q}^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

The model supports two modes of use. In **evaluation mode**, the user supplies a fixed bidding strategy and the model returns the resulting distribution of R^W . In **optimisation mode**, the user searches over admissible strategies using the model as a simulation oracle to identify the strategy that maximises $\mathbb{E}[R^W]$ or another objective, such as a risk-adjusted criterion (e.g. CVaR).

The penalty term $\rho \cdot \delta_t^W$ introduces a fundamental trade-off: bidding a higher quantity v_t^W increases the probability of dispatch and the potential market revenue $\lambda_t \cdot q_t^W$, but also increases exposure to shortfall penalties when realised wind output is low. The optimal strategy must balance these competing effects.

7. Constants

Symbol	Description
$T = 24$	Number of trading intervals per day
$\bar{Q}^{C_k}$	Installed capacity of conventional producer k (MWh)
$\bar{Q}^W$	Nameplate capacity of wind producer (MWh)
$\bar{Q}^S$	Nameplate capacity of solar producer (MWh)
a, b	Baseline offset and amplitude of solar cosine profile (MWh)
$\underline{p}, \bar{p}$	Market-wide bid price floor and cap (\$/MWh)
$\underline{D}$	Demand floor enforcing physical admissibility (MWh)
ρ	Penalty cost per MWh of wind under-delivery (\$/MWh)

8. Uncertain Parameters and Stochastic Variables

Symbol	Description	Distribution
D_t	Hourly market demand	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t ; floored at $\underline{D}$
$\hat{Q}_t^W$	Wind producer realised output	$\mathcal{N}(\mu_W, \sigma_W^2)$, i.i.d. across t ; truncated to $[0, \bar{Q}^W]$
$b_t^{C_k}$	Conventional producer k bid price	$\mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$, i.i.d. across t and across k
b_t^S	Solar producer bid price	$\mathcal{N}(\mu_b^S, \sigma_b^{2,S})$, i.i.d. across t
μ_D, σ_D^2	Mean and variance of hourly demand	Uncertain constants; estimated from historical load data
μ_W, σ_W^2	Mean and variance of wind output	Uncertain constants; estimated from historical wind data

Symbol	Description	Distribution
$\mu_b^{C_k}, \sigma_b^{2,C_k}$	Mean and variance of conventional producer k 's bid price	Uncertain constants; estimated from historical bid data or cost priors
$\mu_b^S, \sigma_b^{2,S}$	Mean and variance of solar producer's bid price	Uncertain constants; estimated from historical bid data

All stochastic variables are assumed mutually independent across agents and across intervals. This precludes, for instance, correlation between wind output and demand, which is a known feature of real electricity markets. This assumption is appropriate for the initial model and may be relaxed in subsequent iterations — for example by replacing the marginal normal distributions with a bivariate normal for $(D_t, \hat{Q}_t^W)$ — without altering the remaining model structure.

The mean and variance of hourly demand, as well as the mean and variance of actual wind-power production are constants. Please refine your specification accordingly.

7. Constants (*Revised*)

Symbol	Description
$T = 24$	Number of trading intervals per day
$\bar{Q}^{C_k}$	Installed capacity of conventional producer k (MWh)
$\bar{Q}^W$	Nameplate capacity of wind producer (MWh)
$\bar{Q}^S$	Nameplate capacity of solar producer (MWh)
a, b	Baseline offset and amplitude of solar cosine profile (MWh)
$\underline{p}, \bar{p}$	Market-wide bid price floor and cap (\$/MWh)
$\underline{D}$	Demand floor enforcing physical admissibility (MWh)
ρ	Penalty cost per MWh of wind under-delivery (\$/MWh)
μ_D, σ_D^2	Mean and variance of hourly market demand (MWh)
μ_W, σ_W^2	Mean and variance of realised wind-power production (MWh)

8. Uncertain Parameters and Stochastic Variables (*Revised*)

Symbol	Description	Distribution
D_t	Hourly market demand	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t ; floored at $\underline{D}$
$\hat{Q}_t^W$	Wind producer realised output	$\mathcal{N}(\mu_W, \sigma_W^2)$, i.i.d. across t ; truncated to $[0, \bar{Q}^W]$
$b_t^{C_k}$	Conventional producer k bid price	$\mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$, i.i.d. across t and across k
b_t^S	Solar producer bid price	$\mathcal{N}(\mu_b^S, \sigma_b^{2,S})$, i.i.d. across t
$\mu_b^{C_k}, \sigma_b^{2,C_k}$	Mean and variance of conventional producer k 's bid price	Uncertain constants; estimated from historical bid data or cost priors
$\mu_b^S, \sigma_b^{2,S}$	Mean and variance of solar producer's bid price	Uncertain constants; estimated from historical bid data

The parameters μ_D, σ_D^2, μ_W , and σ_W^2 are treated as fixed, known constants throughout the simulation, having been relocated from this section to §7. The remaining uncertain parameters — the means and variances governing competitor bid prices — are not directly observable by the wind producer and must be estimated or elicited prior to simulation. All stochastic variables are assumed mutually independent across agents and across intervals.

Various stakeholders approach this problem from different perspectives. Please revise your specification to use agents to represent two perspectives separately and independently: (1) the wind-power producer; and (2) the system regulator. In the revised specification, specify the common environment, including the agents, global constants, uncertain parameters and stochastic variables. For each perspective, specify the agent's key components, including state variables, control variables, processes and objectives.

Agent-Based Model: Day-Ahead Wind-Power Revenue Estimation

Revised Specification

Part I: Common Environment

1. Time Scales

The model operates across two nested time scales.

Scheduling horizon: A single calendar day, subdivided into $T = 24$ sequential hourly trading intervals, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$.

Market clearing: The day-ahead auction is cleared once, simultaneously for all 24 intervals, prior to real-time delivery. Bid submission and market clearing precede physical dispatch; committed quantities become binding obligations subject to penalty upon under-delivery.

2. Agents

Five producers participate as agents, each submitting one price-quantity bid per interval. A sixth, passive entity — the **Market Operator** — clears the market via the merit-order mechanism but makes no strategic decisions.

Agent ID	Technology	Bid behaviour
C_1, C_2, C_3	Conventional (coal)	Stable quantities; stochastic bid prices
S	Solar PV	Deterministic cosine quantity profile; stochastic bid price
W	Wind	User-specified bid quantity and price; stochastic realised output

Two stakeholder perspectives are modelled separately and independently, each observing the same market environment but pursuing distinct objectives:

Perspective	Role
Wind-power producer	Determines bidding strategy to maximise expected net revenue
System regulator	Monitors market outcomes to assess system cost and renewable integration

3. Global Constants

Symbol	Description
$T = 24$	Number of trading intervals per day
$\bar{Q}^{C_k}$	Installed capacity of conventional producer k (MWh)
$\bar{Q}^W$	Nameplate capacity of wind producer (MWh)
$\bar{Q}^S$	Nameplate capacity of solar producer (MWh)
a, b	Baseline offset and amplitude of solar cosine profile (MWh)
$\underline{p}, \bar{p}$	Market-wide bid price floor and cap (\$/MWh)

Symbol	Description
$\underline{D}$	Demand floor enforcing physical admissibility (MWh)
ρ	Penalty cost per MWh of wind under-delivery (\$/MWh)
μ_D, σ_D^2	Mean and variance of hourly market demand (MWh)
μ_W, σ_W^2	Mean and variance of realised wind-power production (MWh)

4. Uncertain Parameters and Stochastic Variables

Symbol	Description	Distribution
D_t	Hourly market demand	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t ; floored at $\underline{D}$
$\hat{Q}_t^W$	Wind producer realised output	$\mathcal{N}(\mu_W, \sigma_W^2)$, i.i.d. across t ; truncated to $[0, \bar{Q}^W]$
$b_t^{C_k}$	Conventional producer k bid price	$\mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$, i.i.d. across t and across k
b_t^S	Solar producer bid price	$\mathcal{N}(\mu_b^S, \sigma_b^{2,S})$, i.i.d. across t
$\mu_b^{C_k}, \sigma_b^{2,C_k}$	Mean and variance of conventional producer k 's bid price	Uncertain constants; estimated from historical bid data or cost priors
$\mu_b^S, \sigma_b^{2,S}$	Mean and variance of solar producer's bid price	Uncertain constants; estimated from historical bid data

All stochastic variables are assumed mutually independent across agents and across intervals.

5. Market Clearing Process

The following process is executed by the Market Operator for each interval t , and is common to both agent perspectives.

Step 1 — Rank bids. Sort all submitted bids in ascending order of bid price. Let σ_t denote this ranking, so that $b_t^{\sigma_t(1)} \leq b_t^{\sigma_t(2)} \leq \dots \leq b_t^{\sigma_t(5)}$.

Step 2 — Identify the marginal price. Find the cheapest combination of bids whose cumulative offered quantity meets or exceeds demand:

$$k^* = \min \left\{ k : \sum_{j=1}^k v_t^{\sigma_t(j)} \geq D_t \right\}$$

The market-clearing price is set at the bid price of the marginal producer:

$$\lambda_t = b_t^{\sigma_t(k^*)}$$

Step 3 — Resolve ties at the marginal price. Let $\mathcal{M}_t$ denote the set of producers whose bid price equals λ_t , and define residual demand at the marginal price as:

$$D_t^{\text{res}} = D_t - \sum_{\{i : b_t^i < \lambda_t\}} v_t^i$$

If the aggregate quantity offered by $\mathcal{M}_t$ exceeds D_t^{res} , producers in $\mathcal{M}_t$ are dispatched according to the fixed priority order $S \succ W \succ C_1 \succ C_2 \succ C_3$. Let $\pi = (\pi_1, \pi_2, \dots)$ be the subsequence of $\mathcal{M}_t$ ordered by priority. Dispatched quantities are:

$$q_t^{\pi_j} = \begin{cases} v_t^{\pi_j} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \\ D_t^{\text{res}} - \sum_{l=1}^{j-1} v_t^{\pi_l} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \leq \sum_{l=1}^j v_t^{\pi_l} \\ 0 & \text{if } \sum_{l=1}^j v_t^{\pi_l} \geq D_t^{\text{res}} \end{cases}$$

Step 4 — Assemble full dispatch. For every producer i :

$$q_t^i = \begin{cases} v_t^i & \text{if } b_t^i < \lambda_t \\ q_t^i \text{ from Step 3} & \text{if } b_t^i = \lambda_t \\ 0 & \text{if } b_t^i > \lambda_t \end{cases}$$

Part II: Wind-Power Producer Perspective

6. State Variables

Symbol	Description	Domain
$\hat{Q}_t^W$	Realised wind output at interval t	$[0, \bar{Q}^W]$ (MWh)
q_t^W	Dispatched quantity at interval t	$[0, v_t^W]$ (MWh)
λ_t	Market-clearing price at interval t	$\mathbb{R}_{\geq 0}$ (\\$/MWh)
δ_t^W	Shortfall at interval t	$\mathbb{R}_{\geq 0}$ (MWh)
r_t^W	Net revenue at interval t	$\mathbb{R}$ (\\$)

7. Control Variables

The wind producer's bid quantity and bid price are free decision variables, specified exogenously by the user for each interval:

$$v_t^W \in [0, \bar{Q}^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

The bid quantity v_t^W represents the quantity committed one day in advance, prior to observing $\hat{Q}_t^W$. These constitute the **primary inputs** to the wind-producer model.

8. Processes

Shortfall calculation. After real-time delivery, realised output is compared against the dispatched commitment:

$$\delta_t^W = \max(0, q_t^W - \hat{Q}_t^W), \quad \forall t \in \mathcal{T}$$

No penalty applies if the producer over-delivers; surplus output beyond q_t^W is uncompensated.

Interval net revenue. The net revenue at each interval combines market revenue and shortfall penalty:

$$r_t^W = \lambda_t \cdot q_t^W - \rho \cdot \delta_t^W, \quad \forall t \in \mathcal{T}$$

Daily net revenue aggregation.

$$R^W = \sum_{t=1}^{24} r_t^W = \sum_{t=1}^{24} (\lambda_t \cdot q_t^W - \rho \cdot \delta_t^W)$$

9. Objective

The wind producer seeks a bidding strategy that maximises expected net daily revenue:

$$\max_{\{b_t^W, v_t^W\}_{t=1}^{24}} \mathbb{E}[R^W] = \mathbb{E}\left[\sum_{t=1}^{24} (\lambda_t \cdot q_t^W - \rho \cdot \delta_t^W)\right]$$

subject to:

$$v_t^W \in [0, \bar{Q}^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

The model supports two modes of use. In **evaluation mode**, the user supplies a fixed bidding strategy and the model returns the distribution of R^W . In **optimisation mode**, the user searches over admissible strategies to identify the pair (b_t^W, v_t^W) that maximises $\mathbb{E}[R^W]$ or a risk-adjusted criterion such as CVaR, with the model serving as a simulation oracle for each candidate strategy. The penalty term $\rho \cdot \delta_t^W$ introduces a fundamental trade-off: committing a higher v_t^W increases potential dispatch revenue but also increases exposure to shortfall penalties when realised wind output is low.

Part III: System Regulator Perspective

10. State Variables

Symbol	Description	Domain
λ_t	Market-clearing price at interval t	$\mathbb{R}_{\geq 0}$ (\\$/MWh)
q_t^i	Dispatched quantity of producer i at interval t	$[0, v_t^i]$ (MWh)
D_t	Aggregate demand at interval t	$\mathbb{R}_{> 0}$ (MWh)
δ_t^W	Wind shortfall at interval t	$\mathbb{R}_{\geq 0}$ (MWh)
E_t	Total system dispatch cost at interval t	$\mathbb{R}_{\geq 0}$ (\\$)
F_t	Renewable fraction of total dispatch at interval t	$[0, 1]$

11. Control Variables

The regulator does not intervene in day-to-day market operations and therefore holds no control variables in the single-day simulation. The regulator is a **passive observer** whose instruments — such as the penalty rate ρ , the priority rule, or the bid price cap $\bar{p}$ — are treated as fixed constants in the current specification. These may become active control variables in an extended multi-scenario formulation aimed at policy optimisation.

12. Processes

System dispatch cost. The total cost of serving demand at interval t , evaluated at the clearing price across all dispatched producers:

$$E_t = \lambda_t \cdot D_t, \quad \forall t \in \mathcal{T}$$

Renewable penetration. The fraction of total dispatched quantity supplied by renewable producers (wind and solar) at interval t :

$$F_t = \frac{q_t^W + q_t^S}{\sum_i q_t^i} = \frac{q_t^W + q_t^S}{D_t}, \quad \forall t \in \mathcal{T}$$

Daily aggregates.

$$E = \sum_{t=1}^{24} E_t, \quad \bar{F} = \frac{1}{24} \sum_{t=1}^{24} F_t, \quad \Delta^W = \sum_{t=1}^{24} \delta_t^W$$

where E is total daily system cost, $\bar{F}$ is mean daily renewable penetration, and Δ^W is total daily wind shortfall.

13. Objective

The regulator monitors market outcomes across the distribution of simulated days to assess three system-level criteria:

System cost efficiency. Minimise the expected total daily cost of serving demand:

$$\min \mathbb{E}[E] = \mathbb{E} \left[\sum_{t=1}^{24} \lambda_t \cdot D_t \right]$$

Renewable integration. Maximise expected mean daily renewable penetration:

$$\max \mathbb{E}[\bar{F}] = \mathbb{E} \left[\frac{1}{24} \sum_{t=1}^{24} \frac{q_t^W + q_t^S}{D_t} \right]$$

Supply reliability. Minimise expected total wind shortfall as a measure of delivery risk:

$$\min \mathbb{E}[\Delta^W] = \mathbb{E} \left[\sum_{t=1}^{24} \delta_t^W \right]$$

These objectives are generally in tension: a wind producer bidding aggressively to maximise revenue may increase shortfall risk, raising system reliability concerns; a high penalty ρ may discourage aggressive bidding and reduce shortfall but may also suppress renewable participation. The regulator's perspective therefore provides the complementary system-level lens through which the investment's broader viability and policy implications can be assessed.

To ensure internal consistency across these perspectives, please compose them into a unified agent-based model. The composed model should: 1. embed each perspective's components, including state variables, control variables, processes and objectives; and 2. enable modular evaluation of each perspective, where other perspectives' controls are treated as exogenous inputs. Please specify the key components of the final model according to the ODD protocol.

Agent-Based Model: Day-Ahead Wind-Power Revenue Estimation

Specification According to the ODD Protocol

OVERVIEW

1. Purpose

The model simulates single-day operations of a day-ahead electricity spot market comprising five producers — three conventional, one solar, and one wind — cleared by a merit-order mechanism. Its primary purpose is to estimate the expected net revenue of the wind-power producer under uncertainty in demand, wind output, and competitor bidding behaviour, thereby supporting an assessment of the economic viability of a wind-power investment.

The model embeds two stakeholder perspectives — the **wind-power producer** and the **system regulator** — as distinct but co-resident agents within a shared market environment. Each perspective exposes its own state variables, control variables, processes, and objectives, while observing the same stochastic realisations and market outcomes. Modular evaluation is enabled by treating each perspective's controls as exogenous inputs when evaluating the other, so that either perspective can be interrogated independently without modifying the shared environment.

2. Entities, State Variables, and Scales

2.1 Time Scales

The model operates across two nested time scales.

Scheduling horizon: A single calendar day subdivided into $T = 24$ sequential hourly trading intervals, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$.

Market clearing: The day-ahead auction is cleared once, simultaneously for all 24 intervals, prior to real-time delivery. Bid submission and market clearing precede physical dispatch; committed quantities become binding obligations subject to penalty upon under-delivery.

2.2 Agents

Six entities populate the model: five producer agents and one passive market operator.

Agent ID	Technology	Role
C_1, C_2, C_3	Conventional (coal)	Submit price-quantity bids; stable quantities, stochastic prices
S	Solar PV	Submits price-quantity bids; deterministic cosine quantity, stochastic price
W	Wind	Submits price-quantity bids; user-specified strategy, stochastic output
$\mathcal{O}$	Market Operator	Clears the market via merit-order; no strategic decisions

Two **stakeholder perspectives** are co-resident within the model, each observing the same environment but holding distinct state variables, controls, and objectives:

Perspective	Primary concern
Wind-power producer ($\mathcal{P}_W$)	Revenue maximisation under output and price uncertainty
System regulator ($\mathcal{P}_R$)	System cost efficiency, renewable integration, and supply reliability

2.3 State Variables

Global market-level state variables (shared across both perspectives):

Symbol	Description	Domain
D_t	Aggregate electricity demand at interval t	$\mathbb{R}_{>0}$ (MWh)
λ_t	Market-clearing price at interval t	$\mathbb{R}_{\geq 0}$ ($\backslash$ \$/MWh)
D_t^{res}	Residual demand at the marginal price at interval t	$\mathbb{R}_{\geq 0}$ (MWh)
q_t^i	Dispatched quantity of producer i at interval t	$[0, v_t^i]$ (MWh)

Wind-producer perspective ($\mathcal{P}_W$) state variables:

Symbol	Description	Domain
$\hat{Q}_t^W$	Realised wind output at interval t	$[0, \bar{Q}^W]$ (MWh)
δ_t^W	Shortfall at interval t	$\mathbb{R}_{\geq 0}$ (MWh)
r_t^W	Net revenue at interval t	$\mathbb{R}$ ($\backslash$ \$)
R^W	Total net daily revenue	$\mathbb{R}$ ($\backslash$ \$)

System regulator perspective ($\mathcal{P}_R$) state variables:

Symbol	Description	Domain
E_t	Total system dispatch cost at interval t	$\mathbb{R}_{\geq 0}$ ($\backslash$ \$)
F_t	Renewable fraction of total dispatch at interval t	$[0, 1]$
E	Total daily system cost	$\mathbb{R}_{\geq 0}$ ($\backslash$ \$)
$\bar{F}$	Mean daily renewable penetration	$[0, 1]$

Symbol	Description	Domain
Δ^W	Total daily wind shortfall	$\mathbb{R}_{\geq 0}$ (MWh)

3. Process Overview and Scheduling

The following sequence is executed once per simulated day, in strict order.

Step	Process	Perspective
1	Draw stochastic realisations	Shared
2	Compute solar bid quantities	Shared
3	Submit all bids to the market operator	Shared
4	Merit-order market clearing	Shared
5	Compute shortfall and net revenue	$\mathcal{P}_W$
6	Compute system cost and renewable penetration	$\mathcal{P}_R$
7	Aggregate daily outputs for both perspectives	Shared

Steps 5 and 6 are logically independent and may be evaluated in either order or in parallel; neither requires outputs from the other. This independence is the structural basis for modular evaluation.

DESIGN CONCEPTS

4. Design Concepts

Emergence. The market-clearing price λ_t and dispatch schedule $\{q_t^i\}$ emerge from the interaction of all five producers' bids and the realised demand draw, via the merit-order mechanism. Neither the price nor the dispatch quantities are prescribed; they are outcomes of the clearing process.

Stochasticity. Five sources of randomness are present: market demand D_t , wind output $\hat{Q}_t^W$, bid prices of each conventional producer $b_t^{C_k}$, and solar bid price b_t^S . All are drawn independently across intervals. The model is evaluated via Monte Carlo simulation over N independent replications to characterise the distributions of all output quantities.

Modularity. The two perspectives are structurally decoupled at the level of state variables, processes, and objectives. The wind producer's controls (b_t^W, v_t^W) are exogenous inputs to the regulator's evaluation; the regulator's policy instruments $(\rho, \bar{p}$, priority rule) are fixed constants in the wind producer's evaluation. Activating one perspective's optimisation does not require modifying the other's specification.

Interaction. Agents interact exclusively through the centralised merit-order clearing mechanism. There is no direct bilateral interaction between producers; each observes only the market-clearing outcome, not the individual bids of rivals.

Objectives. The wind-power producer maximises expected net daily revenue. The system regulator monitors three system-level criteria — cost efficiency, renewable penetration, and supply reliability — which are in general tension with each other and with the wind producer's objective.

Observation. Per simulated day, the model records: $\lambda_t, q_t^i, r_t^W, R^W, \delta_t^W, E_t, F_t$ for all t , and the daily aggregates $E, \bar{F}, \Delta^W$. Across N replications, these are used to estimate expectations, variances, and distributional quantities such as CVaR.

DETAILS

5. Initialisation

Prior to simulation, the following inputs are fixed:

- All global constants (§6) are assigned numerical values.

- The wind producer's bidding strategy $(b_t^W, v_t^W)_{t=1}^{24}$ is specified by the user.
- The regulator's policy instruments $(\rho, \bar{p}, \text{priority rule})$ are assigned their baseline values.
- The number of Monte Carlo replications N is specified.

Each replication begins with a fresh, independent draw of all stochastic variables (§7), with no carry-over of state from prior replications. There is no warm-up period.

6. Global Constants

Symbol	Description
$T = 24$	Number of trading intervals per day
$\bar{Q}^{C_k}$	Installed capacity of conventional producer k (MWh)
$\bar{Q}^W$	Nameplate capacity of wind producer (MWh)
$\bar{Q}^S$	Nameplate capacity of solar producer (MWh)
a, b	Baseline offset and amplitude of solar cosine profile (MWh)
$\underline{p}, \bar{p}$	Market-wide bid price floor and cap (\$/MWh)
$\underline{D}$	Demand floor enforcing physical admissibility (MWh)
ρ	Penalty cost per MWh of wind under-delivery (\$/MWh)
μ_D, σ_D^2	Mean and variance of hourly market demand (MWh)
μ_W, σ_W^2	Mean and variance of realised wind-power production (MWh)

7. Input Data and Uncertain Parameters

The model requires no external time-series inputs. All stochastic quantities are drawn synthetically from the following distributions.

Stochastic variables:

Symbol	Description	Distribution
D_t	Hourly market demand	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t ; floored at $\underline{D}$
$\hat{Q}_t^W$	Wind producer realised output	$\mathcal{N}(\mu_W, \sigma_W^2)$, i.i.d. across t ; truncated to $[0, \bar{Q}^W]$
$b_t^{C_k}$	Conventional producer k bid price	$\mathcal{N}(\mu_b^{C_k}, \sigma_b^{2,C_k})$, i.i.d. across t and k
b_t^S	Solar producer bid price	$\mathcal{N}(\mu_b^S, \sigma_b^{2,S})$, i.i.d. across t

Uncertain parameters (estimated from historical data prior to simulation):

Symbol	Description
$\mu_b^{C_k}, \sigma_b^{2,C_k}$	Mean and variance of conventional producer k 's bid price
$\mu_b^S, \sigma_b^{2,S}$	Mean and variance of solar producer's bid price

All stochastic variables are assumed mutually independent across agents and across intervals.

8. Submodels

8.1 Solar Bid Quantity (Shared)

Executed once per simulated day prior to bid submission:

$$v_t^S = \max\left(0, \, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad \forall t \in \mathcal{T}$$

8.2 Merit-Order Market Clearing (Shared)

Executed by the market operator for each interval t .

Step 1 — Rank bids. Sort all submitted bids in ascending order of bid price. Let σ_t denote this ranking so that $b_t^{\sigma_t(1)} \leq b_t^{\sigma_t(2)} \leq \dots \leq b_t^{\sigma_t(5)}$.

Step 2 — Identify the marginal price. Find the cheapest combination of bids whose cumulative offered quantity meets or exceeds demand:

$$k^* = \min \left\{ k : \sum_{j=1}^k v_t^{\sigma_t(j)} \geq D_t \right\}$$

The market-clearing price is:

$$\lambda_t = b_t^{\sigma_t(k^*)}$$

Step 3 — Resolve ties at the marginal price. Let $\mathcal{M}_t$ denote the set of producers whose bid price equals λ_t , and define residual demand at the marginal price as:

$$D_t^{\text{res}} = D_t - \sum_{\{i : b_t^i < \lambda_t\}} v_t^i$$

If the aggregate quantity offered by $\mathcal{M}_t$ exceeds D_t^{res} , producers in $\mathcal{M}_t$ are dispatched in priority order $S \succ W \succ C_1 \succ C_2 \succ C_3$. Let $\pi = (\pi_1, \pi_2, \dots)$ be the subsequence of $\mathcal{M}_t$ ordered by priority. Dispatched quantities are:

$$q_t^{\pi_j} = \begin{cases} v_t^{\pi_j} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \\ D_t^{\text{res}} - \sum_{l=1}^{j-1} v_t^{\pi_l} & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} < D_t^{\text{res}} \leq \sum_{l=1}^j v_t^{\pi_l} \\ 0 & \text{if } \sum_{l=1}^{j-1} v_t^{\pi_l} \geq D_t^{\text{res}} \end{cases}$$

Step 4 — Assemble full dispatch. For every producer i :

$$q_t^i = \begin{cases} v_t^i & \text{if } b_t^i < \lambda_t \\ q_t^i \text{ from Step 3} & \text{if } b_t^i = \lambda_t \\ 0 & \text{if } b_t^i > \lambda_t \end{cases}$$

8.3 Wind-Producer Perspective ($\mathcal{P}_W$)

Control variables. The wind producer's bid quantity and price are free decision variables, specified exogenously by the user prior to simulation:

$$v_t^W \in [0, \bar{Q}^W], \quad b_t^W \in [\underline{p}, \bar{p}], \quad \forall t \in \mathcal{T}$$

These constitute the primary inputs to $\mathcal{P}_W$ and are treated as fixed exogenous constants from the perspective of $\mathcal{P}_R$.

Shortfall calculation. After market clearing and real-time delivery:

$$\delta_t^W = \max(0, q_t^W - \hat{Q}_t^W), \quad \forall t \in \mathcal{T}$$

Interval net revenue.

$$r_t^W = \lambda_t \cdot q_t^W - \rho \cdot \delta_t^W, \quad \forall t \in \mathcal{T}$$

Daily net revenue aggregation.

$$R^W = \sum_{t=1}^{24} r_t^W$$

Objective. Maximise expected net daily revenue over the user-specified strategy:

$$\max_{\{b_t^W, v_t^W\}_{t=1}^{24}} \mathbb{E}[R^W] = \mathbb{E}\left[\sum_{t=1}^{24} (\lambda_t \cdot q_t^W - \rho \cdot \delta_t^W)\right]$$

subject to $v_t^W \in [0, \bar{Q}^W]$ and $b_t^W \in [\underline{p}, \bar{p}]$ for all $t \in \mathcal{T}$. The model supports two modes: in **evaluation mode**, a fixed strategy is supplied and the model returns the distribution of R^W ; in **optimisation mode**, the model serves as a simulation oracle over the strategy space.

8.4 System Regulator Perspective ($\mathcal{P}_R$)

Control variables. The regulator holds no active controls in the single-day simulation. Its policy instruments ($\rho, \bar{p}$, priority rule) are fixed constants, treated as exogenous inputs from the perspective of $\mathcal{P}_W$. These may be activated as decision variables in an extended multi-scenario policy optimisation.

System dispatch cost.

$$E_t = \lambda_t \cdot D_t, \quad \forall t \in \mathcal{T}$$

Renewable penetration.

$$F_t = \frac{q_t^W + q_t^S}{D_t}, \quad \forall t \in \mathcal{T}$$

Daily aggregates.

$$E = \sum_{t=1}^{24} E_t, \quad \bar{F} = \frac{1}{24} \sum_{t=1}^{24} F_t, \quad \Delta^W = \sum_{t=1}^{24} \delta_t^W$$

Objectives. The regulator monitors three system-level criteria:

$$\min \mathbb{E}[E], \quad \max \mathbb{E}[\bar{F}], \quad \min \mathbb{E}[\Delta^W]$$

These criteria are generally in tension with each other and with the wind producer's revenue objective. A high penalty ρ reduces shortfall risk and supports reliability but may suppress aggressive renewable bidding and reduce penetration; a permissive price cap $\bar{p}$ enables cost recovery for conventional producers but may increase $\mathbb{E}[E]$. The regulator's perspective provides the system-level lens through which the broader implications of the wind producer's bidding strategy — and of the investment itself — can be assessed.

9. Modular Evaluation

The two perspectives are structurally decoupled and may be evaluated independently without modifying the shared environment or any other perspective's specification. The modular evaluation protocol is as follows.

Evaluating $\mathcal{P}_W$ independently. The user supplies $(b_t^W, v_t^W)_{t=1}^{24}$. The regulator's instruments ($\rho, \bar{p}$, priority rule) are held at their assigned constant values. The simulation executes Steps 1–5 and 7 of the process schedule, reporting $\mathbb{E}[R^W]$, $\text{Var}[R^W]$, and related distributional statistics.

Evaluating $\mathcal{P}_R$ independently. The wind producer's strategy $(b_t^W, v_t^W)_{t=1}^{24}$ is held fixed as an exogenous input. The simulation executes Steps 1–4, 6, and 7 of the process schedule, reporting $\mathbb{E}[E]$, $\mathbb{E}[\bar{F}]$, $\mathbb{E}[\Delta^W]$, and related distributional statistics.

Joint evaluation. Both perspectives are evaluated simultaneously under the same set of N Monte Carlo replications, sharing all stochastic draws and market outcomes. This enables direct analysis of the trade-offs between the wind producer's private revenue objective and the regulator's system-level criteria as a function of the bidding strategy $(b_t^W, v_t^W)_{t=1}^{24}$.

Now, please provide a modular Python implementation of the unified model, with each agent implemented as a separate class and a single interface function defined at the end.

The implementation runs cleanly. Here is a summary of how it is structured.

Class structure

Class	Role
ConventionalProducer	Draws i.i.d. normal bid prices; returns fixed capacity as bid quantity
SolarProducer	Computes deterministic cosine quantities; draws i.i.d. normal bid prices
WindProducer	Validates user-supplied bids; draws truncated normal realised output
MarketOperator	Implements the five-step merit-order clearing with priority tie-breaking ($S \succ W \succ C_1 \succ C_2 \succ C_3$)
WindProducerPerspective	Computes shortfall δ_t^W , net interval revenue r_t^W , and daily total R^W
SystemRegulatorPerspective	Computes system cost E_t , renewable fraction F_t , and daily aggregates
SimulationResult	Holds all per-replication arrays and exposes a <code>.summary()</code> method returning $\mathbb{E}[R^W]$, Std, CVaR, $\mathbb{E}[E]$, $\mathbb{E}[\bar{F}]$, and $\mathbb{E}[\Delta^W]$

Interface function

`run_simulation(...)` is the single entry point. It accepts the wind producer's bid strategy as primary inputs, instantiates all agents, runs N Monte Carlo replications, and returns a `SimulationResult`. The two boolean flags `evaluate_wind_perspective` and `evaluate_regulator_perspective` enable modular evaluation: setting either to `False` skips that perspective's computations entirely, leaving the other unaffected.

Modular evaluation

The two perspectives are structurally independent: $\mathcal{P}_W$ uses only `clearing_prices`, `dispatched["W"]`, and `realised_wind`; $\mathcal{P}_R$ uses `clearing_prices`, `demands`, `dispatched["W"]`, `dispatched["S"]`, and the shortfall array. Neither perspective's computations depend on the other's outputs, consistent with the ODD specification.



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